

IoT Driven Air Quality Surveillance and Forecasting

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Abstract— In order to lessen dependency on expensive sensors, this study introduces a low-cost model that makes use of computing power at the edge and the Internet of Things in an effort to develop affordable air quality monitoring and predictive platforms. The Raspberry Pi, also known as a device that provides edges with computing power, applies the Kalman Filter (KF) method to improve low-cost measurements by 27% at the boundary of the device. By combining data with related mistakes, the methodology uses the KF algorithm to predict the level of six pollutants in the air, including SO₂, NO₂, and PM_{2.5}, with lightning speed. The KF procedure's higher predictive ability is shown by comparisons with three traditional prediction techniques: Autonomous Integrated Moving Standard, Increasingly Weighted Move Average, and Simple Moving Average, among others. The root-mean-square error measurement showed a significant decrease of 68.3% on averaged. These results highlight the system's potential to promote intelligent farming and are consistent with its potential for implementation in several domains where precise, immediate time quality of air monitoring is essential.

Keywords— IoT-driven, Air quality surveillance, Forecasting, Edge computing, Kalman Filter, Low-cost sensors, Smart agriculture, Internet of Things (IoT), Real-time prediction and Machine learning algorithms

I. INTRODUCTION

The Internet of Things, or IoT, is a crucial component of the next IT revolution. The term describes a vast system that has been created by connecting several data-gathering gadgets to the Internet. There has been a dramatic increase in the volume of data and connections generated by IoT terminals in recent years, necessitating a redesign of the IoT network to make it easier to analyze and process data in real time. [1] Simultaneously, the infrastructure supporting the IoT is expanding into new areas, such as smart agriculture, smart homes, smart transportation, and more. The network must further decrease the data transmission latency for several specific application scenarios, like security monitoring, automated driving, collecting information about road conditions in real-time, etc. Traditional cellular network architecture lacks the processing and computational capacity needed for the sophisticated IoT, which demands wide coverage and a high number of interactions. Since the cloud-based service and the IoT endpoints are far apart, it is difficult

to provide the immediate information requirements of low-latency technology services.

In order to overcome the IoT development barrier, the proposed edge computing is seen as a crucial enabler. The term "edge computing" describes an open platform that places application core capabilities, data storage, networks, and computations close to the items or data sources themselves. It satisfies critical requirements by offering edge intelligence services. Regarding digitalization in the workplace, some key concerns encompass "real-time commerce, data optimization, application intelligence, privacy, and security." Resembling nerve endings in humans, the characteristics located at the periphery of a network have the ability to independently process simple inputs and transmit the outcomes to the central processing unit in the cloud.

"Smart" agriculture is made feasible by the utilization of sensors and software on computer or mobile platforms to oversee agricultural productivity. This practice is an expansion of the IoT in traditional farming. One of the first and most fundamental steps toward smart agriculture is the installation of an air quality (AQ) monitoring and prediction system that operates in real time. Analyzing the monitoring data allows for the prediction of AQ. Another way of putting it is that the reliability of the monitoring data does impact the prediction's accuracy. Meanwhile, there are AQ values that have been set by the Environment Agency. [4] People need to take the necessary precautions when the air quality reaches a certain level. But it will cause decision-making mistakes if the forecast is wrong. Currently, there are a lot of AQ monitoring methods that rely on high-precision sensors to get their data. Having said that, high-precision sensors tend to be more expensive. There is a compromise between precision and affordability because a whole system requires numerous sensor units. [5] In a traditional IoT-based air pollution surveillance arrangement, the data from the sensing layer needs to be sent to the cloud computing platform of the network component for interpretation and synthesis. Nevertheless, the geographical constraints and limited availability of network infrastructure in China result in the placement of most agricultural fields in remote and challenging locations. The quickness of decision-making is influenced by the uncertainty around the timing of data uploads for monitoring and the analysis of projection conclusions.

According to Fig. 1, the typical cloud computing solutions have a bottleneck when it comes to agricultural application situations.

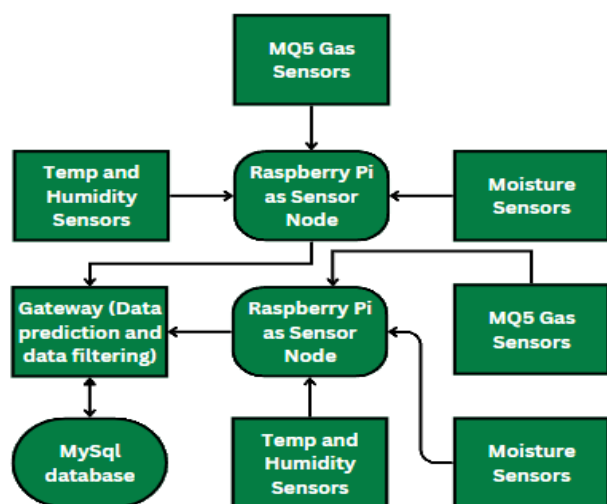


Fig. 1 An Extensive Overview of the Proposed System's Functional Model Revealed

II. LITERATURE REVIEW

A. IoT Edge Technology

The combination of falling CPU and sensors prices has redesigned conventional IoT architectures and accelerated the transition to edge computers. This progression to improve the effectiveness and flexibility of IoT systems has been made necessary by the expansion of networked devices, particularly in areas where computing in the cloud displays inefficiencies. In contrast to cloud computer science, the edge of computing provides observable benefits such increased attention to real-time, short-cycle evaluation of information, which makes it easier to run applications in the cloud that depend on large data analysis. The primary benefits include prompt data collection and evaluation; a reduction in latency by handling data closer to the source; economical local data handling; enhanced programmed runtime efficiency; diminished reliance on cloud services; and the avoidance of possible one-time points of failure. The incorporation of cutting-edge computing into Internet of Things (IoT) platforms has proven beneficial for a variety of applications, ranging from agricultural equipment to driverless systems. Many businesses have taken use of computing at the edges in novel ways. For example,

they have used IoT structures for linked cars, remote control and tracking for intelligent energy control, and fog-enabled embedded technology for monitoring the surroundings. The proliferation of computer systems at the edge is just as important to the emergence of the smart sector as heavy cloud use.

B. System For Assessing Air Efficiency

The convergence of wireless tools, IoT, and tiny device sensors has made it easier to construct air quality (AQ) surveillance networks that are built on conventional IoT foundations. The perception phase is made up of many gadgets with sensors that identify different air properties and compile the information. The data collected by the sensors is processed by the network layer, which then makes decisions and offers insights based on the current air quality conditions. It is made possible by wireless technologies, cloud computing platforms as well and networking administration tools. Users are then provided with pertinent details by the app's layer.

C. Forecasting Model

Predicting concentrations of air pollutants and evaluating air quality in particular locations have been the main foci of the current upsurge in air quality research. It is common practice to employ either traditional statistical methods or ML or multi-model fusion-based approaches for developing prediction models. Approaches centre around "Simple-Moving-Average (SMA), Exponentially-Weighted-Moving-Average (EWMA), and Autoregressive-Integrated-Moving-Average (ARIMA)" are notable in statistical techniques. Due to its simple calculation technique, SMA performs well in immediate prediction, whereas EWMA models provide smoother forecasts that are more in line with current data. Because ARIMA algorithms improve forecast accuracy by filtering non-stationary components, they are often used for AQ forecasting. nevertheless, the understanding and cost-effectiveness of classic statistical techniques are limited when it comes to handling intricate, nonlinear predictions problems. The development of ML techniques, on the other hand, provides greater versatility for intricate predictions. To predict AQ, studies have used multi-model blending and machine learning approaches. The utilization of the ID3 technique in decision trees and ANN-based AQ predicting in models of decision-making has shown noteworthy promise in forecasting levels of airborne contaminants and furnishing timely alerts for heavily contaminated regions.

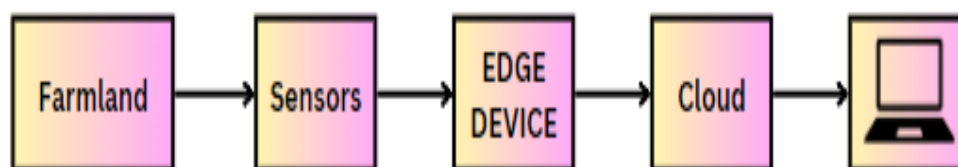


Fig. 2 Architectural Blueprint for Air Quality Monitoring and Predictive Systems within the Framework

The accessibility and compute efficiency of classic statistically based forecasting algorithms make them very effective; yet, their limited attribute expression and incapacity to address intricate, linear forecasting problems limit their intrinsic capabilities. Many research has moved towards ML approaches and multi-model fusion strategies for air quality (AQ) prediction as machine learning (ML) keeps evolving. For example, two different models of decision-making were developed for comparison research that included predicting of the concentrations of PM_{2.5} in two adjacent locations. These models are used to determine the proportion of contaminants in the air by using the algorithm known as ID3 inside a decision tree design. This allows them to provide early warning systems and forecast insights for regions with greater polluted level. Additionally, an Artificial Neural Network (ANN)-based multi-input, multi-output AQ forecasting framework was created, creating a 24-hour predicting model based on the association between the most recent and previous 24-hour concentrations of contaminants data. Sousa uses a mix of principle component-based ANN and linear regression techniques to estimate ozone concentrations. These sophisticated techniques demonstrate the improved predicting powers of machine learning in addressing nuanced AQ forecasts, providing more profound understanding of complex ecological observation.

III. PROPOSED METHODOLOGY

A. The Proposed System Architecture

Both the software and hardware parts are included in the suggested system design. The client interface and clouds storage for data are software-based parts, whereas the Raspberry Pi's, sensors networks, and wireless connectivity module comprise technical pieces. A network of instruments makes it possible to monitor the concentrations of pollutants in real time. The system of sensors sends concentration-related data through the Raspberry Pi after testing. Using this data, the Raspberry Pi, which serves as an edge processing hub, runs the Kalman Filter method, maintaining and going values to forecast ratios for the next seconds. After generating predictions, the Raspberry Pi uses its wireless communication module to send the data over to the clouds. Through client connections and cloud storage, users can access air quality assessments and real-time pollution levels.

B. Hardware

1) Raspberry Pi

It is an efficient, lightweight and extremely efficient bare board system that runs a complete operating system (OS) without the need for additional devices. Raspbian is a Linux operating system built upon Debian that is tailored for Raspberry Pi devices. Because of its integrated quad-core 1.2 GHz 64-bit CPU, the computer known as the Raspberry Pi is a popular option for edge nodes in a variety of Internet of Things-related applications due to its significant processing

capacity. At under \$35, its affordability and easy availability make it the perfect part for a wide range of purposes.

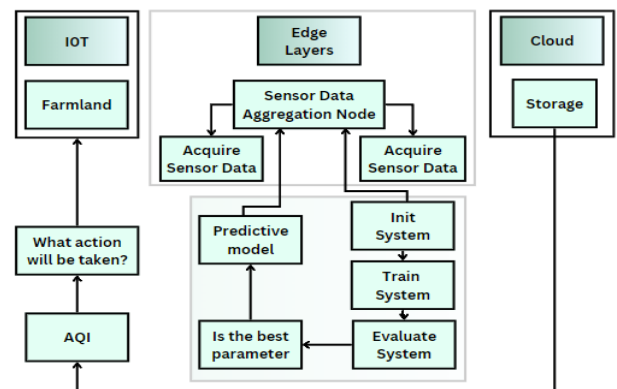


Fig. 3 Incorporating Machine Learning into Edge-to-Cloud Architectures to Enhance System Intelligence

The Raspberry Pi functions as the hub and computational edge unit in a network of sensors due to its ability to process information and ability to compute. Parameter data from a variety of sensors is routinely read by the Raspberry Pi via the matching input pins. A scheduled function is triggered from sleep to constantly run the Kalman Filter mechanism and update the sensor data repeatedly after gathering the monitoring information. The anticipated values for different levels of airborne pollutants at the following timestamp are provided after the computations are complete. The Raspberry Pi then uses the Wi-Fi module to transfer the information from the monitoring and predictions findings into the cloud, which then takes care of storing the information.

2) Sensors

2.1 Laser Dust Sensor ZH03A:

This adaptable, tiny sensor detects dust particles in the air by using laser light scattering. By using distinct channels to differentiate between particle sizes, it produces distinct PM_{2.5} and PM₁₀ percentage readings. The ZH03A sensor has many output possibilities, including online PWM (Pulse Width Modulation), and analogue outputs, along with high reliability, peace of mind, and immediate form responsiveness. The Raspberry Pi receives 24-bit data packets using UART communication, which may be used to extract the PM_{2.5} and PM₁₀ concentration. The ZH03A sensor costs around \$11, to be exact.

2.2 Intelligent Gas Sensor SGA-700:

The signal the process of an analysis of data, the temperature, and wetness correction of these sophisticated gas detection module are well-known. These offer more stability, reduced size, and lower cost. The SGA-700 series has serial port signal capacities as well as direct voltage output inputs in the 0.4–5 V range. You can measure your levels of CO, NO₂, SO₂, and O₃ using the SGA-700 as sensors. These sensors are available for less than \$3 each.

2.3 Wireless Sensor ESP8266:

The Raspberry Pi and the ESP8266 microprocessor, which is well-known for its low power usage and integrated Wi-Fi capabilities, are connected via UART serial connectivity. This enables the Raspberry Pi to communicate with one another and send sensor-monitored information and anticipated pollution findings to the cloud.

The estimated \$75 is the overall cost of implementation for this suggested a solution, which is a substantial reduction from the normal \$750–\$3000 range of current air quality sensors. The suggested method offers more affordable access to every module.

C. Kalman Filter Algorithm

The model of prediction based on classic statistical methods is distinguished by its cheap processing cost and great interpretability. This model works best in situations when the trend of the data is steady since it makes predictions based on past data that has been linearly fitted. Nevertheless, this method is insufficient when different concentrations of air pollution undergo sudden fluctuations, such as when pollutants like PM2.5, PM10, and SO are present during peak traffic hours. Forecasting methods like SMA, EWMA, and ARIMA find it difficult to anticipate these abrupt shifts, despite the data's usefulness.

IV. DISCUSSION

The Kalman Filter technique distinguishes itself from other time series prediction models by its ability to seamlessly include statistical information about process noise and observation error into the prediction process. This enables the program to correct the initial estimation of the model. Hence, the Kalman Filter (KF) utilizes observation data acquired from the sensor network to enhance the precision of predictions, specifically from a prognostic standpoint. The Kalman Filter (KF) can enhance the accuracy of sensors, particularly low-cost and imprecise ones, by rectifying measurement mistakes at the algorithmic level, making it valuable for monitoring purposes. The objective of this study is to assess the efficacy of the Kalman Filter (KF) algorithm in enhancing the precision of sensor observations. In order to accomplish this, a comparative experiment is devised to precisely measure the magnitude of errors. The experiment involves comparing the estimated value of the Kalman Filter (KF) with the actual value of the sensor, using data collected from the official monitoring site.

V. CONCLUSIONS

This research presents a cost-effective approach that utilizes edge computing and the Internet of Things (IoT) to monitor and forecast air quality in smart agriculture. The Raspberry Pi is used in the system's implementation as an edge device that is in charge of carrying out the Kalman Filter technique. This research proposes an innovative approach to air quality monitoring and prediction that makes use of the machine learning algorithm on the edge instead of the traditional cloud-based solution. By lowering data transmission latency, this technique improves real-time decision-making in the agricultural context by mitigating the limits imposed by bandwidth and network connection restrictions. The RPi, the edge device with powerful processing capabilities, is used to run the KF algorithm. This allows six distinct kinds of air pollutants (SO₂, NO₂, CO, O₃, PM_{2.5}, and PM₁₀) to be forecasted in real-time. At the same time, there are issues with the edge computing layer's own monitoring and early warning system. In summary, it is essential to tackle the previously described algorithmic and systemic problems in the future when putting the system into a real-world production setting.

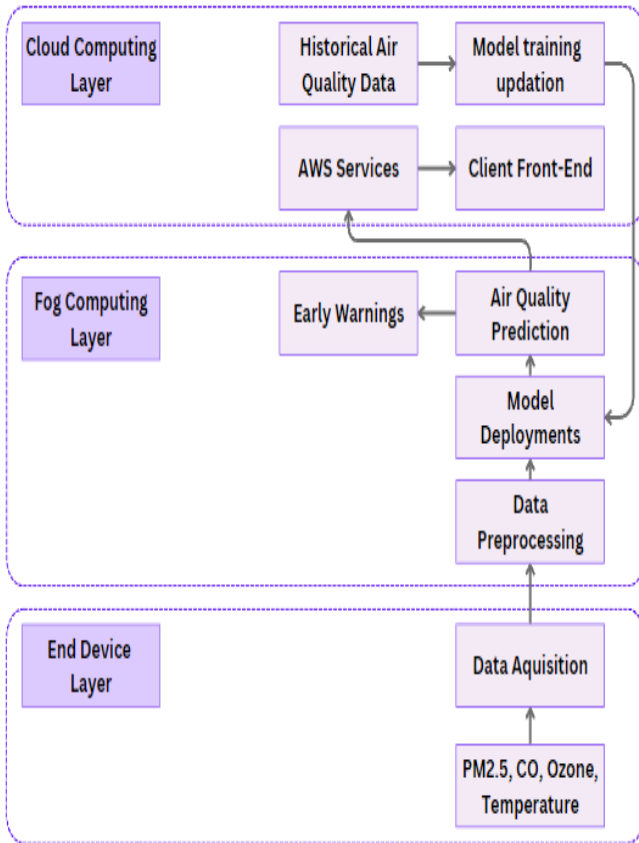


Fig. 4 Realizing the Vision: Strategies and Insights into System Implementation

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